C2lRec: Causal Contrastive Learning for User Cold-start Recommendation with Social Variables

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Embedding-based recommender systems rely on historical interactions to model users, which poses challenges for recommending to new users, known as the user cold-start problem. Some approaches incorporate social networks to deduce preferences based on the social circles of cold-start users to solve the problem of sparse features. However, such methods have difficulty distinguishing between superficial correlations and causal relationships in social behaviors, leading to inaccuracies in predicting user preferences. To address the aforementioned issues, we propose the Causal Contrastive Learning Recommendation (C2lRec) framework. Specifically, we causally model the inference of hidden preferences from the feature and historical behavior of warm users and predict user interactions based on such preferences. The counterfactual inference is

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subsequently performed to intervene and extract interactions from historical behaviors of warm users that influence their preferences, designating as primary causal variables. Additionally, we utilize the primary causal variables from users within the social circle of cold-start users to substitute the missing historical interactions of cold-start users and employ a similar causal modeling approach to uncover hidden preferences as we do with warm users. Finally, we realize causal contrastive learning to enhance the distribution of cold-start users. Extensive experiments conducted on three public datasets demonstrate that the recommendation performance of C2lRec exceeds that of state-of-the-art methods.

CCS Concepts: • **Information systems** → **Recommender systems**; • **Computing methodologies** → **Causal reasoning and diagnostics**;

Additional Key Words and Phrases: Recommender System, Cold-start, Causality Representation Learning, Social Networks

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1 Introduction

Currently, the internet produces vast amounts of data [9], far exceeding the processing and comprehension capabilities of individuals. The advent of recommender systems [8, 23, 25, 57] is specifically designed to tackle the issue of information overload [11], aiding users in navigating the extensive array of information to find content that is relevant and valuable to them. These systems have been widely applied in various fields, such as e-commerce [65, 66], social media [19, 47], and online video and music streaming services [20, 55]. However, most recommender systems predict based on user historical behaviors. Such approaches present a significant challenge when dealing with new users who lack a history of interactions, leading to difficulties in accurately predicting user preferences, i.e., a dilemma commonly referred to as the cold-start problem [1, 6, 36].

To address the cold-start problem, a prevalent approach within the academic community involves leveraging side information of cold-start users [15, 56, 60], such as demographic details including age and income, or utilizing knowledge graphs to refine user embedding representations. However, the effectiveness of side information can be limited due to its inadequacy and potential misalignment with users' actual preferences, often leading to suboptimal outcomes. Another solution employs **Graph Neural Networks (GNNs)** [5, 30, 37], which involves modeling users, items, and their interactions within a graph structure to capture complex relationships and dependencies. Such approaches assist cold-start users in leveraging information from neighboring nodes to enhance their recommendation results. Nevertheless, the inherent data sparsity in recommender systems poses challenges for GNNs, hindering their ability to learn effectively from limited and sparse data. Although some methods have overcome the problem of sparse original graph data through solutions such as sparse matrices and graph reconstruction, the amount of data and the degree of data sparsity in the field of recommendation systems still affect the widespread use of GNNs. While GNN-based recommender systems also use side information, their main advantage is that they can exploit the connection patterns within the graph to learn richer representations, and they do not focus on the use of side information.

Recommender systems that utilize social networks [7, 40, 41] demonstrate significant advantages in resolving the cold-start problem. By utilizing the social connections among users, these systems can deduce user preferences based on their social circles, effectively recommending for new users

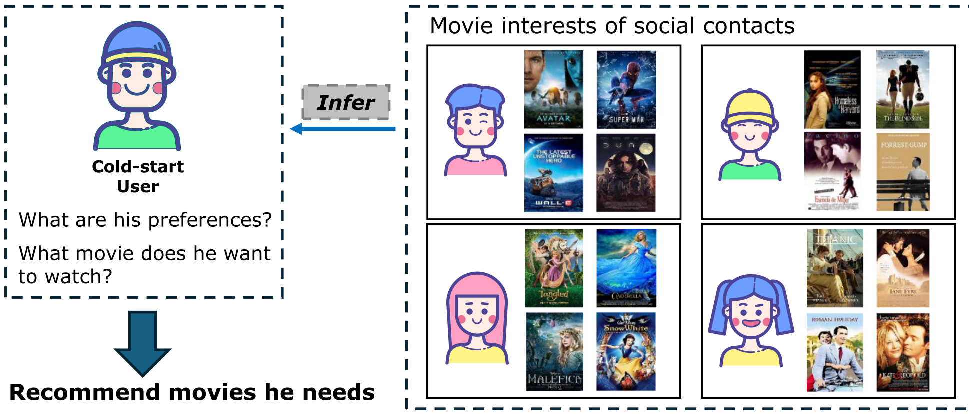


Fig. 1. An illustration of user preference influenced by social context.

or items. As shown in Figure 1, cold-start users lack historical behaviors, yet users within their social circle have interaction records. Such recommender systems can infer the actual preferences of cold-start users based on interactions in their social circle. These methods considerably reduce the dependence on direct historical behavior data from users, enhance the adaptability of the model to cold-start users, and boost user trust and satisfaction with the outcomes of recommendations. However, using social networks in recommender systems also encounters several limitations. First, a user's social connections cannot accurately reflect their true content preferences, resulting in recommendations that fail to meet actual user needs. Moreover, when recommender systems depend on social relationships, they find it challenging to differentiate between superficial correlations and actual causal relationships in social behaviors. For instance, a user engages with certain content out of social obligation or courtesy, interactions that cannot accurately reflect their true interests or preferences.

Recently, numerous researchers have focused on causal representation learning [2, 4, 39, 53, 61]. Traditional machine learning models, which predominantly rely on statistical methods, often assume that data are **Independent and Identically Distributed (IID)**. However, these models typically underperform when test data deviate from the training data distributions, a frequent issue in scenarios where the IID assumption is not met, such as in cold-start situations. Causal representation learning addresses this challenge by extracting data representations that elucidate causal relationships between variables. This foundational understanding enables models to make more stable and accurate predictions amidst changes in distributions or following interventions. This approach enhances model robustness and presents a novel solution to the user cold-start problem.

Based on the discussion above, we propose a **Causal Contrastive Learning framework for user cold-start RECommendation (C2lRec)** to address the user cold-start problem. Causal contrastive learning uses causal inference techniques to infer different types of user preferences and uses contrastive learning to reduce the preference gap between cold-start users and warm users. Building on traditional sequence recommendation methods, we analyze and extract cause-effect factors from the recommendation process. Inspired by the variational auto-encoder [26], for warm users, we extract variables V_1 and V_2 from user features and behavioral sequences, respectively, and predict three types of latent preferences influenced by V_1, influenced by V_2, and jointly influenced by V_1 and V_2. We then predict warm user interactions based on these three types of latent preferences. Subsequently, we employ counterfactual inference to identify the primary causal variables, which are the subset of interactions that most accurately represent user preferences in historical behavior.

Specifically, we hypothesize a scenario in which the user does not interact with a particular item and observe whether there is a significant change in their actual preferences [3, 62]. Meanwhile, we use an attention mechanism to infer the user's long-term preferences and identify primary causal variables for each warm user.

Since users and their social circles often exhibit homophily [34], individuals within the same social group tend to have highly similar preferences in recommender systems [12, 29, 43]. Thus, for cold-start users, we simulate their preferences based on users within their social circle, utilizing the primary causal variables of each user in their social circle. To prevent the excessive influence of individual users' primary causal variables, we first aggregate these variables by individual and then use them to substitute the missing V_2 variable for cold-start users. To address the issue of a significant distribution gap between the preferences of predicted cold-start users and warm users near V_2, the preferences of cold-start users predicted in this manner deviate from warm users, with distributional gaps mainly concentrated around V_2. Therefore, we apply contrastive learning to the preference embeddings influenced by V_2 and jointly by V_1 and V_2, thereby aligning the preference distributions of cold-start users closer to warm users. Extensive experiments on three real-world public datasets further confirm that C2lRec significantly improves recommendation performance for cold-start users without compromising the accuracy of recommendations for warm users.

The main contributions of this article are as follows:

— We designed a framework based on causal contrastive learning (C2lRec) to address the user cold-start problem in recommender systems. By inferring cold-start users' latent preferences based on the overall preferences within their social circles, this framework effectively improves the recommendation performance for cold-start users. Additionally, by combining user features and historical behaviors to infer preferences for warm users, it also significantly enhances recommendation outcomes for warm users.
— C2lRec employs counterfactual reasoning, offering a module based on similarity and long-term user interests to extract primary causal variables. This enables cold-start users to efficiently utilize the historical behaviors of users within their social circles and also provides a degree of interpretability to the model.
— Experiments conducted on three publicly available datasets demonstrate that C2lRec outperforms **State-of-the-Art (SOTA)** models in both warm user and cold-start user scenarios. Ablation studies further confirm the effectiveness of causal inference and contrastive learning.

The organization of this article is as follows. In Section 2, we review other cold-start recommendation methods, recommender systems based on social networks, and recommender systems that utilize causal learning. Section 3 systematically defines sequential recommendations and reconstructs the research problem from a causal perspective. Section 4 provides a detailed description of the C2lRec framework. Extensive experiments are conducted and their results and analyses are presented in Section 5. Finally, Section 6 concludes the article and discusses potential avenues for future work.

2 Related Work

2.1 Cold-start Recommendation

Embedding-based recommender systems utilize machine learning techniques to convert items and users into dense vector representations, commonly known as embeddings, and often face challenges when recommending to cold-start users who lack interaction history with items. Common solutions to address this problem include effectively utilizing attribute features, knowledge graphs, and auxiliary domain information [60]. DropoutNet [49] introduces a dropout that effectively reduces overfitting during training and enhances the robustness of the network. It is a simple yet effective regularization method that is often combined with other cold-start techniques in practical applications to improve recommendation performance during the cold-start phase. MetaEmbedding [35] enhances feature representation by integrating embeddings from multiple information sources. For cold-start items, the generator predicts their initial ID embeddings, which are then used for subsequent training and recommendation. MWUF [67] uses metadata to generate scaling and shifting functions from item features. These functions are applied to transform the features of cold-start items, mapping them into another feature space to improve prediction accuracy. In addition to the aforementioned challenges, effectively utilizing sparse data in cold-start scenarios with limited user–item interactions is also a significant challenge. MetaTL [50] employs a sequential recommendation model that uses few-shot learning to make recommendations for cold-start users. It leverages meta-learning techniques to enhance the accuracy of these recommendations. CDRNP [28] captures the preference correlations between overlapping users and cold-start users through related stochastic processes.

With the widespread use of contrastive learning, a new approach has emerged for addressing the cold-start problem. CLCRec [56] builds on contrastive learning by proposing a new objective function that maximizes the mutual information between feature representations and item collaborative embeddings to address the cold-start problem. Huaye and Chongjun [22] proposed a cold-start recommendation framework based on contrastive learning, which includes three modules: contrastive pair organization, contrastive embedding, and contrastive optimization. This framework allows for collaborative signals to be preserved in content representation for both warm-start and cold-start items.

However, existing methods for addressing these challenges still insufficiently utilize side information, and the systems often suffer from poor generalization capabilities.

2.2 Recommender Systems with Social Network Integration

Social recommender systems differ from traditional recommender systems primarily in two aspects: the recommendation targets and the objectives [10]. Due to changes in feature information and recommendation content in social recommender systems, the associated recommendation methods have also evolved. In attention-based social network recommendation algorithms, the MRAN model [14] proposed by Fu et al. considers both user preferences and user attention. Song et al. [44] developed a neural network model based on dynamic attention graphs. This model combines various user representations for content recommendation using attention mechanisms and GNNs, enhancing the performance of the recommender system.

With the rapid development of deep learning, incorporating deep learning into recommender systems has become a major trend, including in social recommender systems. In the DASRS model [13] proposed by Fan et al., the concept of adversarial network models is introduced, resulting in improved representations of users and items. ARSE [45] focuses on the fact that user preferences consist of both dynamic and static aspects of posted content. To address this, it constructs an RNN-based attention mechanism network model. In the social network domain, research is often conducted by constructing a user–item matrix. However, due to the large volume of data, there is a significant issue of sparsity. Gurini et al. [18] proposed a user recommendation algorithm based on the SVO model, which integrates factors such as sentiment orientation, topic quantity, and objectivity on top of traditional matrix factorization.

In cross-domain recommender systems, social recommender systems play a crucial role. Jian et al. [24] noted that social networks can effectively address the cold-start problem in cross-platform recommender systems. Wang et al. [54] proposed a method called NSCR that seamlessly integrates the user–item domain and the user–user domain, merging information from multiple domains to form cross-domain recommendations and thereby enhancing the performance of the recommender system.

However, among the numerous recommender systems in social networks, most researchers focus on the various feature information and recommended content of the social network, with little consideration given to causal aspects. As a result, it is challenging to trace the reasons behind recommendations and the interpretability is relatively poor.

2.3 Recommender System Based on Causal Learning

Mainstream recommender systems primarily learn from data correlations to determine user preferences. However, the real world is driven by causal relationships rather than mere correlations. As a result, researchers in recommender systems are increasingly using causal inference to improve performance by addressing data bias, handling missingness and noise, and going beyond mere accuracy [17].

To address data bias, such as popularity bias or exposure bias, some existing work tackles these issues through backdoor adjustment methods. PDA [64] eliminates the confounding popularity bias during model training and adjusts the recommendation scores through causal intervention to account for the expected popularity bias, thereby leveraging the popularity bias to improve recommendation accuracy. DecRS [51] employs approximate backdoor adjustment techniques to handle confounding factors and dynamically optimize recommendation performance. In handling missing data and noise, data missingness issues include limited user data collection and the causal impact of the recommendation model on the system. In the extreme case, this can even lead to data noise problems. In the first case, counterfactual reasoning can be used to generate uncollected data as a supplement to existing data, thereby addressing the issue of data missingness. CIRS [16] combines causal inference with offline reinforcement learning to infer user satisfaction. By using a causal user model, it mitigates overexposure effects and provides unbiased counterfactual rewards. In the second case, causal models such as Inverse Probability Weighting can be used to estimate the causal effects of the recommendation model. Sato et al. [38] proposed an unbiased causal effect learning framework for recommendations, which utilizes inverse propensity score techniques for unbiased estimation and reduces variance through empirical risk minimization. In the area of causal inference-based recommender system modeling beyond accuracy, considerations extend beyond just recommendation precision.

Moreover, these systems can also focus on metrics such as diversity, interpretability, controllability, and fairness. Models designed based on causal graphs are inherently controllable. High diversity can be achieved by controlling the model to avoid tradeoffs. Wang et al. [52] proposed the UCRS recommendation prototype, which flexibly mitigates filter bubbles through four types of user controls and the UCI framework. Fair recommendations can be achieved by controlling the model to ensure equitable recommendations for specific user groups. Zhang et al. [63] proposed the CBDF method, which adjusts delayed feedback rewards through counterfactual reasoning to ensure that reward estimates are statistically unbiased and have sub-linear regret bounds.

Although existing research has addressed data biases and handled missing data and noise through causal inference methods, improving recommender system performance, these studies generally do not consider the cold-start user scenario. Integrating causal inference with strategies for handling cold-start users is an important direction for future research, aiming to enhance the adaptability and accuracy of recommender systems for new user scenarios.

3 Problem Formulation

In this section, we will model sequential recommendations incorporating user features. Additionally, we will model cold-start recommendations that incorporate features and social networks from a causal perspective. We will begin with the fundamental concepts and mathematical notations addressed in this article.

3.1 Sequential Recommendations Incorporating User Features

Sequential recommendation models accurately simulate user behavior sequences by leveraging historical user actions, thus predicting items that users are likely to interact with in the future and enhancing overall recommendation performance. However, the pronounced reliance of these models on historical user behaviors leads to suboptimal performance in cold-start scenarios, where new users lack historical data. To address this challenge, we enhance user modeling by incorporating user features, which significantly improves model performance.

In the recommendations section of this article, the set of users is represented by U and the set of items by I. Users are designated as $u_i \in U, i = 1, 2, ..., N$, where N represents the total number of users. The historical interactions of user u_i are expressed as $S^{u_i} = \{i_1^{u_i}, i_2^{u_i}, ..., i_{(n-1)}^{u_i}, i_n^{u_i}\}$, with n indicating the count of items with which the user has interacted. The term $i_j^{u_i}(1 <= j <= n)$ specifies the jth item engaged by user u_i. The features associated with user u_i are denoted by F^{u_i}. The objective of sequential recommendations that incorporate user features is to predict the next item $y^{u_i} = i_n^{u_i}$ that user u_i is likely to engage with, based on $x^{u_i} = \{i_1^{u_i}, i_2^{u_i}, ..., i_{(n-1)}^{u_i}\}$ in conjunction with F^{u_i}. This prediction can be mathematically formulated as the probability of user u_i interacting with all items.

$$p\left(y^{u_i} = i | x^{u_i}, F^{u_i}\right),\ i \in I. \tag{1}$$

Specifically, the objective in the context of the recommendation is to predict the interaction probability between the current user and all items based on the user's historical behavior x^{u_j} and user features F^{u_j}.

It is particularly noteworthy that for the input of user u_i, $x^{u_i} = \{i_1^{u_i}, i_2^{u_i}, ..., i_{(n-1)}^{u_i}\}$, the embedding requires encoding of each $i_j^{u_i}$ individually to facilitate the generation of social variables and the implementation of counterfactual verification subsequently.

3.2 Causal Cold-start Recommendation Incorporating Social Networks

From a causal perspective, we represent the user features and historical behaviors influencing user preferences as observable variables, while user preferences themselves are modeled as latent variables. The causal relationships underlying the overall recommendations are depicted in Figure 1.

In this model, V_1 and V_2, respectively, denote the observable variables corresponding to user features and historical behaviors. H_1 represents latent user preferences influenced exclusively by V_1, H_2 captures the latent preferences influenced solely by V_2, and $H_{(1,2)}$ embodies the latent preferences influenced by both V_1 and V_2; G symbolizes the items with which users interact, which are also the items that should be recommended to the users.

For cold-start users, who have minimal historical behavior information, the previously discussed causal modeling is ineffective. Social networks often demonstrate "homophily," suggesting from a recommender system perspective that a user's circle of friends tends to share similar tastes, interests, or consumption behaviors. Consequently, it is likely that a user's behavioral patterns closely mirror those of their friends within their social network.

For the reasons mentioned previously, in the case of cold-start users, we replace the user's historical behaviors with a set of key factors from social networks (predominantly influential behaviors within friends' sequences) as V_2. We utilize counterfactual inference for further validation, and we will provide detailed explanations of this approach in subsequent chapters.

The objective at this stage is to provide overall social preferences for cold-start users and predict the items they prefer based on social preferences. The objective function can be represented as follows:

$$p\left(y^{u_j} = i | x^{u_j}, F^{u_j}, Soc^{u_j}\right), \, i \in I, \tag{2}$$

where Soc^{u_j} represents the social preference of user u_j.

4 Methodology

In this section, we introduce a causal inference-based approach to user cold-start recommendations, aimed at enhancing recommendation performance by leveraging features and social networks of cold-start users. The objective is to overcome the shortcomings of current cold-start recommender systems that depend solely on user features or popular items, resulting in poor generalization capabilities, weak generalization capabilities. Moreover, causal recommendations provide a degree of interpretability, clarifying the principal factors influencing the recommendations.

4.1 Overview

Our approach adopts a causal perspective to investigate the relationships between user features, historical behaviors, and user preferences, thereby enhancing conventional sequential recommendations. Initially, we identify and subsequently validate the primary influencing factors within user behavior sequences using counterfactual inference. We then exploit similarities in behavioral patterns within the same social circles, extracting primary influencing factors from the historical behaviors of other users in a cold-start user's social circle to substitute for the cold-start user's missing historical behavior. Ultimately, we ensure the convergence of the user cold-start recommendation model, by utilizing contrastive learning, guided by the model developed for warm users.

As shown in Figure 2, the complete model includes three distinct components: the warm user causal recommendation module, the primary influencing factor extraction and validation module, and the cold-start user social causal recommendation module, detailed as follows:

— *Warm User Causal Recommendation Module*: This module integrates user features and historical behaviors to identify hidden user preferences from a causal perspective, thereby providing more generalized and accurate recommendations. Specifically, we employ an attention mechanism to encode the user feature sequence and historical behavior sequence, extracting the information of the variables V_1 and V_2. We then use the encoders E_1, E_2, and $E_{(1,2)}$ to capture V_1 and V_2 and predict the hidden preferences H_1, H_2, and $H_{(1,2)}$. Finally, we combine the recommendation loss with the **Evidence Lower Bound (ELBO)** to ensure that the model achieves high recommendation accuracy while maintaining close alignment between the posterior distribution and the true distribution. Overall, this module aims to strike a balance between recommendation accuracy and regularization of the latent space.

— *Primary Causal Variables Extraction Module*: This module is designed to identify the primary causal variables within user behavior sequences. For the input sequence of user u_j, expressed as $x^{u_j} = \{i_1^{u_j}, i_2^{u_j}, ..., i_{(n-1)}^{u_j}\}$, we calculate the similarity of each item to $i_n^{u_j}$. Furthermore, we engage in counterfactual inference for user u_j by hypothesizing scenarios where the user has not interacted with item combinations exhibiting high similarity and analyze the resulting discrepancies in predictions. Ultimately, for each user, we identify the item combination exerting primary causal variables, deemed as the principal factor shaping their historical behavior. We assert that the prediction of a user's preferences is primarily influenced by these primary causal variables, which play a significant role in the recommendation process.

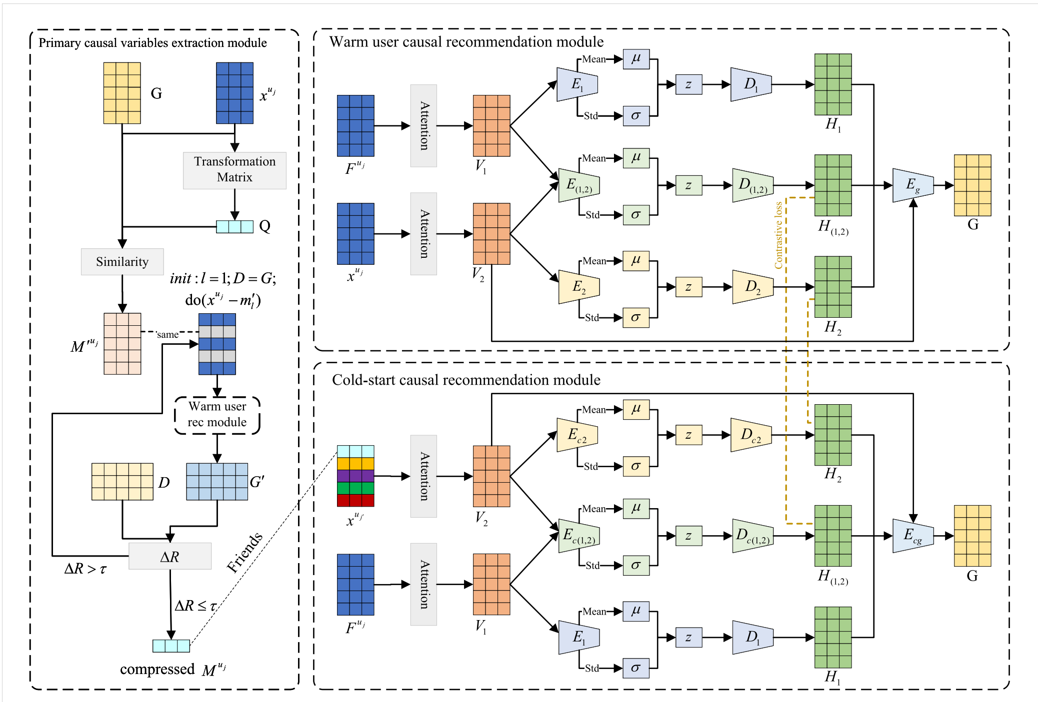


Fig. 2. The overall architecture of C2IRec.

— *Cold-start Causal Recommendation Module*: This module synthesizes user features and social network preferences to forecast the preferences of cold-start users, thus delivering high-accuracy recommendations for such users. For cold-start user $u_{j'}$, we identify and aggregate primary causal variables from their social circle on an individual basis to prevent the predominance of overly lengthy item combinations attributable to any single user. Following this, the cold-start social embedding $t^{u_{j'}}$ supersedes their initial input $x^{u_{j'}}$ as the user's V_2. The model integrates preferences H_2 and $H_{(1,2)}$, utilizing contrastive learning methods to reduce disparities with the preferences of warm users, effectively preventing biased treatment of cold-start users. Ultimately, we calculate the probabilities of interactions between $u_{j'}$ and all potential items i.

In subsequent sections, we will provide detailed introductions to each module within the framework.

4.2 Warm User Causal Recommendation Module

The objective of this module is to dissect three categories of user preferences from a causal standpoint: H_1, H_2, and $H_{(1,2)}$. In this context, H_1 represents latent user preferences solely affected by V_1, while H_2 captures those solely influenced by V_2, and $H_{(1,2)}$ represents preferences concurrently influenced by both V_1 and V_2. V_1 and V_2 correspond to observable user characteristic variables and user historical behavior variables, respectively. Subsequently, the analysis extends to examining user interaction behavior G in light of H_1, H_2, and $H_{(1,2)}$.

As traditional sequence models often fail to sustain long-term dependencies with increasing sequence lengths, we utilizes an attention mechanism to encode the original sequences before undertaking causal inference. Specifically, for the input behavior sequence of user u_j, denoted as $x^{u_j} = \{i_1^{u_j}, i_2^{u_j}, ..., i_{(n-1)}^{u_j}\}$, where $i_k^{u_j}(1 \leq k \leq n-1)$ represents the embedding of the kth interaction item of user u_j, we will refer to these embeddings concisely as i_k. Learnable transformation matrices M_1, M_2, and M_3 are introduced to perform the following transformations on i_k:

$$\alpha_1^k = M_1 i_k,\ \alpha_2^k = M_2 i_k,\ \alpha_3^k = M_3 i_k. \tag{3}$$

Similar to the attention mechanism, we use the obtained α_1^k and α_2^k to calculate the similarity between different items and use α_3^k to preserve the original information of the current item. To calculate the similarity $\rho_{<k,l>}$ between the embeddings of two items, i_k and i_l, we utilize α_1^k and α_2^l obtained from the preceding formula. The exact calculation is detailed as follows:

$$\rho_{<k,l>} = \frac{\exp(\alpha_1^k \cdot \alpha_2^l/\sqrt{n-1})}{\sum_{m=1}^{n-1} \exp(\alpha_1^k \cdot \alpha_2^m/\sqrt{n-1})}. \tag{4}$$

Given the variability in the lengths of historical behavior sequences among different users, the magnitude of the inner product between α_1^k and α_2^l escalates exponentially with dimensionality. To mitigate the impact of dimensionality on model training, it is imperative to implement additional normalization procedures. The previously acquired α_3^k retains the original information for the item at position k. By applying a similarity-based weighting to this information, x^{u_j} can be transformed into the user preference variable $V_2^{u_j} = \{a_1^{u_j}, a_2^{u_j}, ..., a_{(n-1)}^{u_j}\}$, where the computation formula for $a_k^{u_j}(1 \leq k \leq n-1)$ is specified as follows:

$$a_k^{u_j} = \sum_{m=1}^{n-1} \rho_{<k,m>} \cdot \alpha_3^k. \tag{5}$$

Evidently, each element $a_k^{u_j}(1 \leq k \leq n-1)$ within the user preference variable $V_2^{u_j}$ comprehensively integrates all pertinent information from the historical behavior of users, weighted by similarity. The implementation of only three learnable transformation matrices, M_1, M_2, and M_3, ensures that incorporating the attention mechanism does not exacerbate the computational demands of model training. Subsequently, by employing the same method and utilizing the three learnable transformation matrices M_4, M_5, and M_6, the user features F^{u_j} of u_j can be mapped to the user feature variable V_1.

Drawing on the foundational work of prior researchers [53, 61], we assume that user preferences follow factorized Gaussian priors. We utilize encoders E_1, E_2, and $E_{(1,2)}$ to enable the transformation of V_1 and V_2 into H_1, H_2, and $H_{(1,2)}$, respectively. These encoders are represented by the functions Φ_1, Φ_2, and $\Phi_{(1,2)}$. Consequently, the distributions for H_1, H_2, and $H_{(1,2)}$ are specified as follows:

$$\begin{cases} H_1 \sim \mathcal{N}(\mu_1(V_1), \mathrm{diag}\{\sigma_1^2(V_1)\}) \\ H_2 \sim \mathcal{N}(\mu_2(V_2), \mathrm{diag}\{\sigma_2^2(V_2)\}) \\ H_{(1,2)} \sim \mathcal{N}(\mu_{(1,2)}(V_1, V_2), \mathrm{diag}\{\sigma_{(1,2)}^2(V_1, V_2)\}), \end{cases} \tag{6}$$

where, $\mu_1(V_1)$ signifies the estimated mean of the Gaussian distribution derived from the function $\Phi_1(V_1)$, while $\mathrm{diag}\{\sigma_1^2(V_1)\}$ represents the estimated diagonal covariance of this distribution. Consequently, H_1 conforms to a Gaussian distribution characterized by a mean of $\mu_1(V_1)$ and a variance of $\mathrm{diag}\{\sigma_1^2(V_1)\}$, effectively capturing its influence on user preferences. Likewise, H_2 is described by a Gaussian distribution with mean $\mu_2(V_2)$ and variance $\mathrm{diag}\{\sigma_2^2(V_2)\}$. The mean and variance of $H_{(1,2)}$ are collaboratively computed from both V_1 and V_2, demonstrating their synergistic impact on user preferences.

For the user interaction item G, we utilize the encoder E_g to integrate the historical behavior variable V_2 with the user preferences H_1, H_2, and $H_{(1,2)}$ for prediction, represented by the function Φ_g. For the sake of convenience, $\mathscr{H}$ is used instead of $H_1, H_2, H_{(1,2)}$ hereinafter. Accordingly, the user interaction item G is expressed using the following formula:

$$G = \Phi_g(V_2, \mathscr{H}). \tag{7}$$

For a batch of size N, the users within the batch can be represented as $U_B = \{u_1, u_2 \ldots, u_b\}$. The ground truth for the recommendations is given by $y = a_n^{u_j}$. The partial loss function for the recommendation can be expressed as follows:

$$\mathscr{L}_{rec} = -\frac{1}{N} \sum_{u_j \in U_B} \left\{ a_n^{u_j} - \ln\left(\sum_{l=1}^{n-1} \exp(a_l^{u_j}) \right) \right\}. \tag{8}$$

From a causal perspective, our objective is to minimize the divergence between the posterior distribution inferred by the encoder E_g and the true distribution, while simultaneously maximizing the log-likelihood of the observed data. We achieve this by optimizing the model through the minimization of the negative ELBO, which is defined as follows:

$$\text{ELBO} = \mathbb{E}_{q(\mathscr{H}|V_1,V_2)} \ln(p(G|\mathscr{H})) - \text{KL}(q(\mathscr{H}|V_1, V_2)||p(\mathscr{H})), \tag{9}$$

where, $q(H_1, H_2, H_{(1,2)}|V_1, V_2)$ represents the probability distribution of user preferences as predicted by the encoders E_1, E_2, and $E_{(1,2)}$, while $p(G|H_1, H_2, H_{(1,2)})$ denotes the user interaction item inferred based on these predicted preferences. The term $p(H_1, H_2, H_{(1,2)})$ refers to the true distribution of user preferences. The first term corresponds to the log-likelihood function of the encoder E_g, ensuring the model's accuracy in assessing user interaction items. The second term is the Kullback-Leibler divergence, which measures the difference between the posterior distribution inferred by the encoders E_1, E_2, and $E_{(1,2)}$ and the underlying prior distribution.

Given the unknown joint probability in the first term, calculating the first term in the ELBO poses a challenge. To address this, we employ the Monte Carlo method, drawing N samples from E_1, E_2, and $E_{(1,2)}$. For each sampled variable, we individually evaluate the log-likelihood function of G, resulting in a Monte Carlo estimate of the log-likelihood. The calculation is expressed as follows:

$$\mathbb{E}_{q(\mathscr{H}|V_1,V_2)} \ln(p(G|\mathscr{H})) \approx \frac{1}{N} \sum_{l=1}^{N} \ln(p(G|\mathscr{H})). \tag{10}$$

For the entire module, the loss function can be expressed as follows:

$$\mathscr{L}_1 = \mathscr{L}_{rec} - \theta\, \text{ELBO}, \tag{11}$$

where, θ is a hyper-parameter that controls the weight of ELBO, allowing the model to balance between recommendation accuracy (through $\mathscr{L}_{rec}$) and regularization of the latent space (through ELBO). By minimizing $\mathscr{L}_1$, the model can predict the hidden preferences of user while modeling the probability distribution of user interactions.

4.3 Primary Causal Variables Extraction Module

The objective of this module is to extract the primary causal variables in each user behavior sequence following the completion of training in the preceding module. Initially, it is essential to calculate the similarity between items in historical interactions and the recommendation ground truth, as items exhibiting higher similarity are more likely to constitute the primary influencing factors. However, this approach encounters substantial challenges. For instance, consider a scenario where a user purchases two computers; based on the principle of similarity, the first computer acquired should ostensibly serve as the primary influencing factor for acquiring the second. Nevertheless, in practice, once a user possesses one computer, he will unlikely to procure a second, hence the first should not be considered as the primary influencing factor.

Consequently, we need to extract user long-term interests, focusing more on user interest in the similarity calculations rather than solely on item similarity. Initially, the input behavior sequence for user u_j, $x^{u_j} = \{i_1^{u_j}, i_2^{u_j}, ..., i_{(n-1)}^{u_j}\}$, is formatted as a matrix $X \in \mathbb{R}^{(n-1)\times d}$, where d represents the dimensionality of the items in the sequence. The item $i_n^{u_j}$ is then depicted as $\mathcal{Y} \in \mathbb{R}^{1\times d}$. Learnable parameter matrices $D_1 \in \mathbb{R}^{\frac{d}{2}\times d}$ and $D_2 \in \mathbb{R}^{1\times \frac{d}{2}}$ are constructed, using the following formula to extract the user's long-term interest vector:

$$Q = \text{softmax}(D_2 \tanh(D_d X^T))^T X, Q \in \mathbb{R}^{1\times d}. \tag{12}$$

This formula methodically reduces dimensions to capture the intricate interrelationships and dependencies among various behavioral items in the input sequence. It simplifies these complex relationships into a singular vector, Q, which encapsulates the user's comprehensive long-term interests.

The subsequent step involves calculating the similarity between each $i_k^{u_j} (1 \leq k \leq n-1)$ and both Q and $\mathcal{Y}$. In our experiments, we utilize cosine similarity $\cos_{\text{sim}}(a, b)$ to assess the degree of similarity between two vectors. The specific calculation formula is presented as follows:

$$\cos_{\text{sim}}(a, b) = \frac{a \cdot b}{||a||_2 \times ||b||_2}, \tag{13}$$

where $a \cdot b$ represents the dot product of vectors a and b, while $||a||_2$ denotes the Euclidean norm of vector a. Subsequently, we calculate the similarity between the user interest vector Q and each item in the user's historical behavior, producing the score vector $Score_Q = \{s_1^Q, s_2^Q, ..., s_{(n-1)}^Q\}$. The calculation formula is presented as follows:

$$s_k^Q = \cos_{\text{sim}}(i_k, Q),\ 1 \leq k \leq n-1. \tag{14}$$

Similarly, we calculate the similarity between $\mathcal{Y}$ and each item in the historical behaviors of the user, resulting in the score vector $Score_{\mathcal{Y}} = \{s_1^{\mathcal{Y}}, s_2^{\mathcal{Y}}, ..., s_{(n-1)}^{\mathcal{Y}}\}$. The calculation formula is as follows:

$$s_k^{\mathcal{Y}} = \cos_{\text{sim}}(i_k, \mathcal{Y}),\ 1 \leq k \leq n-1. \tag{15}$$

Finally, the scores from both measures are weighted to obtain the final item scores $Score = \{s_1, s_2, ..., s_{(n-1)}\}$:

$$Score = \jmath Score_Q + (1 - \jmath) Score_{\mathcal{Y}}. \tag{16}$$

For the $n-1$ items in the sequence, we select the highest-scoring $\lfloor \frac{n-1}{2} \rfloor$ items as candidate primary causal variables. In training the parameter matrices $D_1 \in \mathbb{R}^{\frac{d}{2}\times d}$ and $D_2 \in \mathbb{R}^{1\times \frac{d}{2}}$, we aim to minimize the gap with the user preference vector from the previous module, $V_2^{u_j} = \{a_1^{u_j}, a_2^{u_j}, ..., a_{(n-1)}^{u_j}\}$, to derive a long-term interest vector that better reflects user preferences. The loss function is specified as follows:

$$L_2 = 1 - \sum_{l=1}^{n-1} \cos_{\text{sim}}(Q, a_l^{u_j}). \tag{17}$$

While employing interest scores to identify primary causal variables is simple, it can cause the recommender system to excessively depend on a user's historical behaviors. This dependence overlooks latent interests and contextual factors, potentially intensifying data biases and compromising the fairness of the extraction process. To address these issues, we introduce counterfactual inference to enhance the extraction of primary causal variables from the user behavior sequence.

Algorithm 1: Counterfactual Inference for Identifying Primary Causal Variables

Input: $x^{u_j} = \{i_1^{u_j}, i_2^{u_j}, ..., i_{(n-1)}^{u_j}\}$, $M'^{u_j} = \{m'_1, m'_2, ..., m'_{\lfloor \frac{n-1}{2} \rfloor}\}$,
precision loss threshold τ, initial recommendation accuracy R_{org}.
Output: Primary causal variables sequence M^{u_j}.

```
Initialize M^{u_j} = ∅.
Initialize l = 1.
while l ≤ ⌊(n−1)/2⌋ do
    Apply intervention: do(x^{u_j} − m'_l).
    Compute new recommendation accuracy R_new.
    Calculate precision loss: ΔR = R_org − R_new.
    if ΔR > τ then
        Update R_org = R_new.
        Add m'_l to M^{u_j}.
        Increment l = l + 1.
    else
        Break
    end if
end while
Return M^{u_j}.
```

During the similarity computation process, we generate a sequence of candidate primary causal variables, sorted in descending order according to *Score*, resulting in the sequence $M'^{u_j} = \{m'_1, m'_2, ..., m'_{\lfloor \frac{n-1}{2} \rfloor}\}$. Each $m'_l, 1 \leq l \leq \lfloor \frac{n-1}{2} \rfloor$, corresponds to an element within the behavior sequence $x^{u_j} = \{i_1^{u_j}, i_2^{u_j}, ..., i_{(n-1)}^{u_j}\}$.

First, initialize the sequence of primary causal variables $M^{u_j} = \varnothing$, establish the precision loss threshold τ, and utilize the model from the first module to compute the initial recommendation accuracy R_{org}. Traverse the sequence $M'^{u_j} = \{m'_1, m'_2, ..., m'_{\lfloor \frac{n-1}{2} \rfloor}\}$, beginning from $l = 1$, and sequentially apply the following do-operator for intervention:

$$\mathrm{do}(x^{u_j} - m'_l), 1 \leq l \leq \lfloor \frac{n-1}{2} \rfloor. \tag{18}$$

The operation in the formula represents the removal of the item corresponding to m'_l from the user's historical behavior sequence x^{u_j}, effectively simulating a counterfactual scenario: what would the recommendation results differ if the user had not interacted with the item m'_l. Following this modification of the historical behavior for inference, a new recommendation accuracy, R_{new}, is determined. The item m'_l is subsequently added to the primary causal variables sequence M^{u_j}, and the precision loss is then calculated:

$$\Delta R = R_{org} - R_{new}, \tag{19}$$

where ΔR represents the precision loss resulting from the intervention, which is then compared to the precision loss threshold τ. If $\Delta R > \tau$, R_{org} is updated to R_{new}, l is incremented by one, and the do-operator is repeatedly applied for further interventions. If $\Delta R \leq \tau$, then M^{u_j} is identified as the primary causal variables sequence for user u_j. The algorithm for the counterfactual inference process is detailed in Algorithm 1.

4.4 Cold-start Causal Recommendation Module

This module integrates user features and social network preferences of cold-start users to predict their preferences and utilizes contrastive learning to minimize the gap in preference distributions between cold-start users and warm users. During the network training process, we simulate a cold-start scenario by using data from warm user u_j, excluding their historical behavior.

For cold-start users $u_{j'}$, their friend sequence is denoted as $Friends^{u_{j'}} = \{u_1, u_2, ..., u_h\}$. Our objective is to deduce their historical behavior preferences based on the interaction items of their friends. Initially, we extract a friend u_l from $Friends^{u_{j'}}$. In the second module, we ascertain M^{u_l}, which identifies the primary causal variables in the behavior sequence of user u_l that impact their preferences.

Given that the length of the primary causal variables sequence varies among friends, directly using M_{u_j} to represent the preferences of friend u_j can lead to an overrepresentation of those with longer sequences, thereby biasing the prediction of user interaction preferences. Therefore, it is crucial to compress the vector M^{u_l} by averaging all items to derive the vector m^{u_j}, which epitomizes the preference items of friend u_j. Subsequently, $x^{u_{j'}} = \{m_1^{u_{j'}}, m_2^{u_{j'}}, ..., m_{\not{f}}^{u_{j'}}\}$ substitutes the user's input behavior sequence $x^{u_j} = \{i_1^{u_j}, i_2^{u_j}, ..., i_{(n-1)}^{u_j}\}$, and acquire variables V_1 and V_2, H_2 and $H_{(1,2)}$.

Regarding network structure, this module utilizes the same architecture as the first module, thereby sharing only the E_1 parameter along with the transformation matrices M_4, M_5, and M_6. The encoders relevant to this module are identified as E_{c2}, $E_{c(1,2)}$, and E_{cg}, while the transformation matrices are designated as M_{c1}, M_{c2}, and M_{c3}. Given that there is no variance in H_1 preferences between warm and cold users, and a substantial distribution gap arises in the H_2 and $H_{(1,2)}$ preferences post-training, it becomes imperative to apply contrastive learning to H_2 and $H_{(1,2)}$ to minimize their differences with warm users. For this, we use the H_2^+ and $H_{(1,2)}^+$ preferences predicted by the first module for warm user u_j as positive samples in contrastive learning. Preferences from other cold-start users are used as negative samples, designated as H_2^- and $H_{(1,2)}^-$.

For the preferences H_2 and $H_{(1,2)}$ of cold-start users, we utilize a contrastive learning method to minimize the discrepancy with the positive samples H_2^+ and $H_{(1,2)}^+$, thus reducing their distributional difference from warm users. Simultaneously, we maximize the distance from the negative samples H_2^- and $H_{(1,2)}^-$ to ensure the network accurately extracts the user's true preferences. The contrastive loss function is defined as follows, where $\cos_{\text{sim}}(\cdot)$ is succinctly denoted as $C(\cdot)$:

$$\mathscr{L}_c = -\frac{1}{N}\sum_{l=1}^{N}\log\left(\frac{\exp\{(C(H_2^{u_{j'}}, H_2^{u_j})/\xi) + (C(H_{(1,2)}^{u_{j'}}, H_{(1,2)}^{u_j})/\xi\}}{\sum_{t=1}^{N}\{\exp(C(H_2^{u_{j'}}, H_2^{u_{p'}}/\xi) + \exp(C(H_{(1,2)}^{u_{j'}}, H_{(1,2)}^{u_{p'}}/\xi)\}}\right), 1 \leq p \leq h, p \neq j, \quad (20)$$

where ξ serves as a hyper-parameter that regulates the model's sensitivity to the similarity between positive and negative samples. Excessively high values of ξ will lead to insufficient distinction between positive and negative samples, impairing the effectiveness of contrastive learning; conversely, overly low values of ξ could cause the model to focus excessively on negative samples, thereby impeding convergence. The formula demonstrates that the primary aim of this loss function is to reduce the distribution gap between the preferences of cold-start and warm users, while concurrently enhancing the model's capacity to differentiate between various cold-start users, thus averting homogenization.

The module's output is defined as $y = a_n^{u_j}$. The formula for the overall loss function of the module is as follows:

$$\mathscr{L}_3 = \mathscr{L}_{rec} - \theta' \text{ ELBO} + \lambda\mathscr{L}_c. \quad (21)$$

By minimizing $\mathscr{L}_3$, the model can predict the user's hidden preferences while modeling the probability distribution of user interactions, effectively reducing the preference distribution gap between cold-start users and warm users, thus further improving the recommendation performance for cold-start users.

4.5 Training Strategy

The training process of C2lRec is structured into three distinct stages, each corresponding to a different module. Given that the training for the second and third stages depends on the completion of the preceding stage—specifically, the second stage training depends on the first, and the third on the second—the sequence of training must be adhered to strictly and cannot be conducted concurrently. During the training of subsequent modules, the earlier modules must continue to update their parameters.

The warm causal recommendation module utilizes data from warm users and employs supervised learning to train the model, with the aim of enhancing recommendation accuracy for these users. Parameters for encoders E_1, E_2, $E_{(1,2)}$, and E_g, as well as the transformation matrices M_1, M_2, and M_3, are updated in accordance with the loss function $\mathscr{L}_1$.

The primary causal variables extraction module leverages the network trained during the initial phase to infer the variable $V_2^{u_j}$ for user u_j, using it as training data. The module is designed to ensure that Q accurately represents the user's long-term interests, with transformation matrices D_1 and D_2 being updated based on the loss function $\mathscr{L}_2$.

The cold-start causal recommendation module, although sharing the same network structure as the warm causal recommendation module, only shares the E_1 parameter with it. Similar to the warm module, it updates the encoders E_1, E_2, $E_{(1,2)}$, and E_g using both recommendation loss and ELBO. Differing from the warm module, this module introduces a contrastive loss to bridge the distribution gap between the preferences of cold-start and warm users. Importantly, the contrastive learning component utilizes self-supervised learning, eliminating the need for additional data as all data are generated internally by the model itself.

The overall training process is shown in Algorithm 2.

5 Experiment

In this section, we conducted comprehensive experiments on three real-world public datasets to evaluate our proposed C2lRec framework. Specifically, the experiments were designed to address the following research questions:

RQ1: Does the proposed C2lRec framework outperform SOTA methods?
RQ2: How effective is each component of the C2lRec framework?
RQ3: Is there a gap in preference prediction between cold-start and warm users within the framework?
RQ4: How do hyper-parameters, such as θ and $\mathscr{s}$, influence our model's performance?

5.1 Experimental Settings

5.1.1 Datasets. Our methodology requires social data to accurately infer the preferences of cold-start users. Therefore, the datasets we selected include user ratings and the social connections among users. To assess the effectiveness of the C2lRec, experiments were conducted on three real-world public datasets: Epinions [33, 46], Douban [32], and Yelp.

Algorithm 2: C2lRec Training Strategy

Input: x^u , y^u and $Friends^u$ for $u \in U$, learning rate lr, hyperparameters $\theta, \varsigma, \theta', \xi, \lambda$.
Output: Global model parameters $\mathscr{W}$ (consists of $\mathscr{W}_1$, $\mathscr{W}_2$ and $\mathscr{W}_3$.).

Remove user-item interaction of some users to simulate cold-start users.
/* Warm causal recommendation module. */
for $i \leftarrow 1 : E_1$ (number of pre-train epochs) **do**
    for $j \leftarrow 1 : B_1$ (number of pre-train batch size) **do**
        Calculate $\mathscr{L}_{rec}$ by Equation (7).
        Calculate ELBO by Equation (8).
        Calculate $\mathscr{L}_1$ by Equation (10).
        Apply Adam optimizer to $\mathscr{L}_1$.
        Perform back-propagation to $\mathscr{L}_1$ getting gradients $\mathscr{G}_1$.
        Update $\mathscr{W}_1$ (Parameters for encoders E_1, E_2, $E_{(1,2)}$, and E_g, and transformation matrices $M_{1\sim6}$) based on $\mathscr{G}_1$.
    end for
end for
/* Primary causal variables extraction module. */
for $i \leftarrow 1 : E_1$ (number of pre-train epochs) **do**
    for $j \leftarrow 1 : B_1$ (number of pre-train batch size) **do**
        Calculate $V_2^u, u \in U_j$ (Represents the user in the current batch) by $M_{1\sim3}$.
        Calculate $\mathscr{L}_2$ by Equation (16).
        Apply Adam optimizer to $\mathscr{L}_2$.
        Perform back-propagation to $\mathscr{L}_2$ getting gradients $\mathscr{G}_2$.
        Update $\mathscr{W}_2$ (transformation matrices D_1 and D_2) based on $\mathscr{G}_2$.
    end for
end for
/* Cold-start causal recommendation module. */
for $i \leftarrow 1 : E_3$ (number of fine-train epochs) **do**
    for $j \leftarrow 1 : B_3$ (number of fine-train batch size) **do**
        Calculate $\mathscr{L}_{rec}$ by Equation (7).
        Calculate ELBO by Equation (8).
        Calculate $\mathscr{L}_c$ by Equation (19).
        Calculate $\mathscr{L}_3$ by Equation (20).
        Apply Adam optimizer to $\mathscr{L}_3$.
        Perform back-propagation to $\mathscr{L}_3$ getting gradients $\mathscr{G}_3$.
        Update $\mathscr{W}_3$ (Parameters for encoders E_1, E_{c2}, $E_{c(1,2)}$, E_{cg}, and transformation matrices $M_{c1\sim c3}$, $M_{4\sim6}$) based on $\mathscr{G}_3$.
    end for
end for
return $\mathscr{W}$

— *Epinions*: This dataset is collected from an online consumer review Web site. It encompasses user ratings for various items and trust relationships among users. We refined the data by assessing user interaction frequency and the number of friends, subsequently eliminating inactive users and associated items. Moreover, trust relationships were transformed into a social network, where users with mutual trust are considered friends. In total, 21,094 users and 89,537 items were included.

Table 1. Statistics of Experimental Datasets

Dataset	#User	#Item	#Interaction	Sparsity
Epinions	21,094	89,537	419,542	99.97%
Douban	2,732	16,967	513,934	98.89%
Yelp	17,264	20,266	383,764	99.89%

— *Douban*: Douban is a well-regarded social platform in China, particularly known for Douban Movies which provides users with up-to-date movie information. The dataset contains user ratings for movies along with their social connections. Similar to the Epinions dataset, we cleaned less active users. Eventually, we gathered data on 2,732 users and 16,967 items.
— *Yelp*: This dataset comprises a subset of user and review data from the Yelp platform, initially curated for the Yelp dataset challenge. It features user reviews of businesses, as well as their friends lists. Following the same procedure as with the other datasets, we excluded less active users and unrelated items and converted the friends lists into a social network. A total of 17,264 users and 20,266 items were compiled.

For all datasets, 30% of the users were designated for the test set; within this subset, interaction records for 15% of the users were removed to simulate cold-start scenarios, while the remaining 15% were left unchanged. The other 70% of the users were divided into a training set and a validation set using an 8:2 ratio. Table 1 provides a detailed statistical summary of the datasets.

5.1.2 Evaluation Metrics. To evaluate the performance of the C2lRec framework, we employ the **Hit Ratio (HR@K)** and **Normalized Discounted Cumulative Gain (NDCG@K)** as our evaluation metrics. We consider the last item in each user's interaction sequence as the one with which the user actually interacted, and we estimate the interaction probabilities for each item, ranking them in descending order to form our predictions.

HR@K represents whether the actual user interactions occur within the top K items of the predictions. The calculation formula is as follows:

$$\text{HR@}K = \frac{1}{N_{test}} \sum_{l=1}^{N_{test}} \zeta(g^l \in \hat{G}_K), \tag{22}$$

where N_{test} denotes the number of users in the test set, g^l represents the item with which the lth user actually interacted, and $\hat{G}_K$ refers to the set of the top K items predicted by the model that the user prefer. $\zeta(bool)$ is an indicator function, defined as $\zeta(\text{True}) = 1$ and $\zeta(\text{False}) = 0$. NDCG@K assesses the quality of these rankings, and the formula for its calculation is as follows:

$$\text{NDCG@}K = \frac{1}{N_{test}} \sum_{l=1}^{N_{test}} \sum_{k=1}^{K} \frac{\log_2 2 \cdot \zeta(g^l \in \hat{G}_K[k])}{\log_2(k+1)}. \tag{23}$$

In our experiments, K is set to 20. Setting $K = 20$ for evaluating a recommendation system's performance with metrics such as HR@K and NDCG@K can be justified based on several factors. First, $K = 20$ is often chosen to reflect the typical depth of user engagement in browsing recommendations, providing a realistic assessment of the system's ability to capture user interest. Additionally, a single K value simplifies experimental design and comparisons with other studies, maintaining clarity and focus in the analysis. Lastly, using $K = 20$ aligns with established benchmarks in current studies [21, 42], facilitating direct comparison across different studies and ensuring that the findings are relevant to common user scenarios.

5.1.3 Baselines. To evaluate the performance of the C2lRec framework, we compared it with the following baseline models:

— *DropoutNet* [49]: DropoutNet leverages dropout techniques to train neural networks, facilitating effective recommendations even without user or item preference data.
— *Heater* [68]: Heater offers personalized transformation functions for various users and items, solving the issue that a uniform transformation function fails to capture the diversity and complexity of auxiliary information. This approach enhances effective model learning and mitigates typical errors in converting auxiliary information to collaborative filtering.
— *MeLU* [27]: MeLU utilizes the **Model-Agnostic Meta-Learning (MAML)** algorithm, quickly adapting and predicting user preferences following interactions with a few items. Its core strategy involves swiftly learning a user's unique preferences from a limited consumption history, thus facilitating personalized recommendations.
— *MetaHIN* [31]: MetaHIN integrates **Heterogeneous Information Networks (HIN)** with meta-learning methods, improving recommendation performance by creating rich semantic contexts and swiftly adapting to new tasks.
— *CLCRec* [56]: CLCRec, grounded in contrastive learning, seeks to enhance item representation learning by maximizing the mutual information between item content features and collaborative signals. It features a co-adaptive meta-learner for simultaneous semantic and task adaptation, optimizing the meta-learning process and quickly adapting to new users or items under limited interaction data conditions.
— *ColdNAS* [58]: ColdNAS employs a hyper-network to directly map a user's interaction history to user-specific parameters that adjust the prediction model. This method stands out by searching for suitable modulation structures, including both functions and locations, which enhances recommendation performance beyond the capabilities of traditional methods with fixed modulation functions and predetermined modulation locations.
— *M2EU* [59]: M2EU enhances the representation of cold-start users by integrating information from similar users and employing a variance-based attention mechanism to aggregate embeddings of similar users. Moreover, M2EU implements different neural layers at the rating level, introducing a weight-sharing strategy to prevent inadequate parameter learning at certain rating levels.

5.1.4 Experiment Setup. In our experiments, the C2lRec framework was implemented using PyTorch, and the computations were conducted on a high-performance server equipped with two NVIDIA RTX 3090 GPUs, ensuring efficient parallel processing and training. For both the baseline models and C2lRec, the embedding dimension was standardized at 64 to ensure consistent comparisons across all methods. Users with fewer than four interactions in their historical behavior sequence and those with fewer than four friends were excluded from the dataset to avoid sparsity issues, which could negatively affect model training and performance evaluation.

Regarding hyper-parameter configurations, we configured θ and θ' at 0.7, ξ at 0.9, λ as 0.6, and $\mathfrak{s}$ at 0.5. The mini-batch gradient descent algorithm was employed as the optimization strategy, with the batch size configured to 500, allowing the model to efficiently process large datasets while maintaining stable convergence.

We used the Adam optimizer, which is known for its adaptive learning capabilities, with a learning rate set at 0.0015 to facilitate gradual yet steady updates to the model parameters. To prevent overfitting and enhance generalization, early stopping was applied based on the NDCG@K performance on the validation set, where training was terminated if performance did not improve over a set number of epochs.

Table 2. Warm User Recommendation Performance Compared to the Baseline

Method	Epinions		Douban		Yelp	
	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})
DropoutNet	2.73	7.30	0.47	0.82	6.92	7.42
Heater	3.04	8.17	0.62	0.75	6.34	8.34
MeLU	2.91	7.13	0.54	0.63	4.57	6.15
MetaHIN	2.85	7.84	0.73	<u>0.84</u>	5.14	7.85
CLCRec	3.17	8.47	0.31	0.49	7.34	8.13
ColdNAS	3.08	<u>8.61</u>	0.81	0.74	<u>7.87</u>	<u>8.43</u>
M2EU	<u>3.19</u>	8.23	<u>0.84</u>	0.72	6.92	8.34
C2lRec	**3.44**	**8.97**	**0.93**	**0.95**	**8.32**	**8.94**
improve	7.83%	4.18%	10.71%	13.09%	5.71%	6.04%

Table 3. Cold-start User Recommendation Performance Compared to the Baseline

Method	Epinions		Douban		Yelp	
	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})
DropoutNet	1.42	5.41	0.32	0.57	3.72	4.13
Heater	1.96	6.84	0.44	0.52	6.08	4.02
MeLU	1.57	7.14	0.48	0.68	5.27	4.83
MetaHIN	1.78	6.29	0.56	0.65	5.31	4.79
CLCRec	1.95	<u>7.21</u>	0.21	0.35	5.86	<u>4.85</u>
ColdNAS	<u>2.13</u>	7.19	0.63	0.71	4.25	4.76
M2EU	2.06	7.03	<u>0.68</u>	<u>0.74</u>	<u>6.13</u>	4.62
C2lRec	**2.36**	**7.84**	**0.79**	**0.87**	**6.58**	**5.18**
improve	10.79%	8.73%	16.17%	17.56%	7.34%	6.80%

To ensure a fair and consistent evaluation, all baseline models were implemented using publicly available source codes provided by the authors of the respective models. The hyper-parameters and model architectures for these baselines were configured according to the descriptions in the original publications, ensuring that each model was evaluated under optimal conditions. Additionally, extensive experiments were conducted to validate the performance of C2lRec in comparison to these SOTA methods, with a focus on cold-start recommendation performance.

5.2 Performance Comparison (RQ1)

To address RQ1, we carried out an extensive performance assessment of the C2lRec framework alongside SOTA cold-start recommendation models on three datasets. Tables 2 and 3 illustrate the performance of these models in warm user and cold-start user scenarios, measuring outcomes in terms of HR@K and NDCG. The underlined sections highlight the best baseline models for each metric in the dataset, while the bolded parts represent the metrics of C2lRec. Additionally, we present the relative improvements of C2lRec over these baseline models. The results indicate that the proposed method surpasses SOTA methods across all evaluation metrics on all datasets. From the data in the tables, we can derive the following insights:

— Both tables clearly indicate that DropoutNet underperforms in both warm user and cold-start user scenarios. By employing dropout techniques to simulate the absence of user or item data, DropoutNet can make recommendations even without extensive historical interaction data.

However, its failure to fully utilize side information about users results in subpar performance in cold-start scenarios compared to other methods. In warm user scenarios, where there is abundant interaction data, DropoutNet also lags behind methods that leverage this data more effectively.

— We observe that MeLU significantly underperforms on the Yelp dataset relative to the other two datasets. This could be attributed to MeLU's MAML algorithm, which allows for rapid adaptation to new users with minimal interactions, as evidenced in the Douban dataset. Although MeLU is effective at learning long-term user preferences, its forte—rapid generalization from minimal data—may not be advantageous in data-abundant environments such as Yelp.

— Analysis of the data in Table 3 shows that CLCRec demonstrates unexpected precision in the cold-start scenario of the Douban dataset. This may stem from CLCRec's use of contrastive learning to improve initial item or user representations, which demands high-quality and sufficient initial data. The significant imbalance in item content descriptions in the Douban dataset poses challenges for the model to develop robust embeddings capable of accurately capturing user preferences in cold-start situations, leading to suboptimal recommendation outcomes.

— ColdNAS exhibits consistently high performance across all scenarios. Its hyper-network structure is adept at rapidly adapting to specific requirements of new users, dynamically searching for suitable model parameters, which makes it exceptionally well-suited for managing the complexities of cold-start user recommendations. Furthermore, in environments with extensive user data, ColdNAS excels at customizing model parameters, demonstrating a remarkable capacity for flexible adaptation.

C2lRec demonstrates significant improvements in recommendation performance over baseline models across both warm and cold-start scenarios. In the warm user setting, C2lRec consistently outperformed all baselines across Epinions, Douban, and Yelp datasets, achieving average improvements of 8.08% in HR and 7.10% in NDCG. These enhancements reflect C2lRec's ability to effectively model user preferences even with limited historical data. For cold-start scenarios, C2lRec's advantages are even more pronounced, with average gains of 11.43% in HR and 11.03% in NDCG, showcasing its strength in mitigating the cold-start challenge. These results highlight the robustness and efficiency of the C2lRec framework, which leverages social variables and causal contrastive learning to deliver more accurate and reliable recommendations than SOTA baseline models across diverse datasets.

The exceptional performance of the C2lRec framework is attributed to the deployment of several key technologies: First, C2lRec employs counterfactual inference to thoroughly analyze and validate the primary causal variables that influence user decisions. This method more effectively discerns and leverages the variables that drive user choices instead of simply correlating superficial data. Second, the model capitalizes on behavioral patterns within social networks to furnish viable behavioral characteristics for cold-start users who lack extensive historical data. Such analysis of social networks significantly boosts the model's prowess in making recommendations for new users. Lastly, the incorporation of contrastive learning ensures uniformity in recommendations between cold-start scenarios and warm user interactions, thus bolstering the model's overall stability and precision. The combined use of these strategies enables C2lRec to sustain robust performance in complex and variable recommendation settings.

5.3 Ablation Study (RQ2)

To validate the effectiveness of the components within the model, we conducted ablation studies for both warm user and cold-start user scenarios. These experiments involved removing specific

Table 4. Ablation Experiment in Warm User Scenario

Method	Epinions		Douban		Yelp	
	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})
C2lRec	3.44	8.97	0.93	0.95	8.32	8.94
C2lRec-NoAttn	3.12	8.41	0.74	0.89	8.27	8.72
C2lRec-NoELBO	1.46	4.95	0.53	0.62	5.41	4.03

Table 5. Ablation Experiment in Cold-start User Scenario

Method	Epinions		Douban		Yelp	
	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})	HR(e^{-2})	NDCG(e^{-2})
C2lRec	2.36	7.84	0.79	0.87	6.58	5.18
C2lRec-NoQ	2.15	7.42	0.73	0.82	6.41	5.02
C2lRec-NoLc	1.77	5.39	0.37	0.52	4.94	2.73
C2lRec-NoELBO$_c$	1.45	4.10	0.42	0.36	3.73	3.61

components from the original C2lRec framework to construct model variants. We then performed performance evaluations across three datasets, presenting the results using the metrics HR@K and NDCG@K (where $K = 20$).

5.3.1 Warm User Scenario. In the warm user scenario, the performance of the C2lRec framework is primarily influenced by the attention transformation matrices outlined in formula (2) and the ELBO described in formula (8). Consequently, we designed the following two variants of C2lRec for evaluation:

— C2lRec-NoAttn: To assess the importance of the attention mechanism within the model, we removed the transformation matrices $M_{1\sim3}$ indicated in formula (3) and $M_{4\sim6}$ mentioned in later sections from the original C2lRec. Instead, we directly used the embedding of users' historical behavior as the V_1 variable for input.
— C2lRec-NoELBO: To examine the model's precision in estimating user preferences, we set θ in formula (10) to 0 and trained the model solely using the recommendation loss.

From the data presented in Table 4, it is clear that removing the attention mechanism leads to a decrease in HR and NDCG across all datasets. This finding highlights the essential role of the attention mechanism in precisely capturing both long-term and short-term user interests, which is particularly noticeable with the notable drops in performance in the Epinions and Yelp datasets. This emphasizes the value of the attention mechanism in processing extensive user behavior data to improve the model's predictive accuracy and relevance. Additionally, the absence of the ELBO for causal extraction significantly lowers model performance. For example, in the Epinions dataset, HR reduced from 0.0344 to 0.0146, and NDCG declined from 0.0897 to 0.0495. Such marked declines illustrate the critical function of ELBO in helping the model deduce users' latent preferences. ELBO facilitates the model in efficiently extracting the causal relationships between user preferences and their characteristics, as well as between user preferences and their historical behaviors without sacrificing the accuracy of the recommendations, by ensuring a tight alignment between the posterior and true distributions. Across all datasets, the complete C2lRec model, which includes both the attention mechanism and ELBO, surpasses versions that lack these integral components, proving that these features are indispensable for enhancing the performance of recommender systems.

5.3.2 Cold-start User Scenario. In the cold-start user scenario, the primary factors influencing the C2lRec framework include the user's long-term interest vector outlined in formula (11), the contrastive loss calculated in formula (19), and the ELBO as detailed in formula (20). Consequently, we developed the following three variants of C2lRec for this scenario:

— C2lRec-NoQ: This variant tests the importance of the long-term interest vector Q. We removed the transformation matrices that normally integrate the similarities between items and the user's long-term interests, which are critical for precisely capturing interactions that genuinely reflect user preferences. Instead, we set $Score = Score_{\mathcal{Y}}$ as per formula (15) and recalculated the similarity to assess how the absence of these matrices affects the model's performance.
— C2lRec-NoLc: Considering the potential significant distributional differences between cold-start and warm users, this variant employs contrastive learning to minimize these gaps. By setting λ in formula (20) to 0, we removed the contrastive loss from the loss function and retrained the cold-start causal recommendation module to evaluate the impact of omitting this component.
— C2lRec-NoELBO$_c$: This variant tests the model's accuracy in estimating cold-start user preferences by setting θ' in formula (20) to zero. The model is trained using only the recommendation and contrastive losses, allowing us to determine the specific contributions of the ELBO to the model's effectiveness.

This setup allows us to isolate and evaluate the individual contributions of key components within the C2lRec framework, enhancing our understanding of their impact on the model's performance in cold-start scenarios.

Compared to the complete C2lRec model, the C2lRec-NoQ variant exhibits a slight reduction in HR and NDCG across all datasets. This highlights the critical role of the long-term interest vector in accurately reflecting user preferences, emphasizing the substantial influence of analyzing user historical interactions on the recommender system's performance.

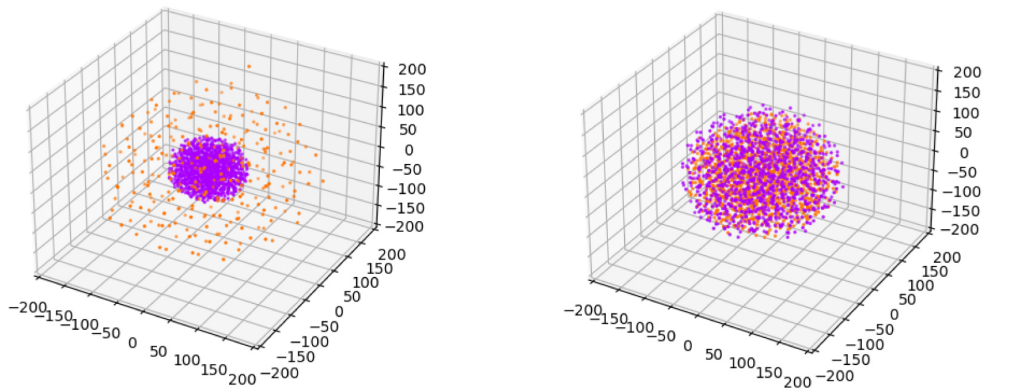
Across all datasets, eliminating contrastive learning results in a marked decline in performance, notably in the NDCG metric. Contrastive learning within the model aims to minimize the disparities between cold-start users and warm users. In the absence of the contrastive loss, the model loses a vital mechanism for fine-tuning and optimizing new users' feature representations, thereby increasing susceptibility to errors in predicting new user preferences. This issue is particularly crucial in cold-start scenarios, where the limited user data significantly reduces the model's generalization capability and adaptability, severely impacting the accuracy and relevance of the recommendations.

Moreover, removing ELBO deprives the model of its ability to deduce latent user preferences, leading to inaccurate predictions for cold-start user preferences and a substantial decline in recommendation performance. The lack of these components directly affects the model's essential ability to address cold-start problems, particularly in understanding and adapting to new user preferences.

Across all datasets, the full C2lRec model surpasses the variants lacking these critical components, demonstrating that these features are vital for optimizing recommendation performance for cold-start users.

5.4 Visualization Study (RQ3)

To address RQ3, we analyzed the preference vectors of cold-start and warm users following various training approaches and conducted a visual analysis. Utilizing t-SNE [48], we reduced the dimensionality of the user behavior preference vectors H_2, detailed in the METHODOLOGY section, to three dimensions. These data points were visually represented, as illustrated in Figure 3. This approach allows a clearer comparison of the distinct behavioral patterns between cold-start and warm users under different training conditions.



(a) Distribution of H_2 between cold-start users and warm users after $\mathscr{L}_{rec}$ training.

(b) Distribution of H_2 between cold-start users and warm users after Complete training.

Fig. 3. Comparison between H_2 between cold-start users (purple) and warm users (orange).

In scenarios where only recommendation training is applied (as depicted in Figure 3(a)), it is evident that the preference vectors of cold-start users (represented by purple points) are relatively clustered in the space, while those of warm users (orange points) are broadly dispersed. This observation suggests that without the integration of causal extraction or contrastive learning techniques, the limited historical behavior of cold-start users leads to less diverse preference vectors. Conversely, warm users, benefiting from a richer historical dataset, display more diverse and scattered behavioral preferences.

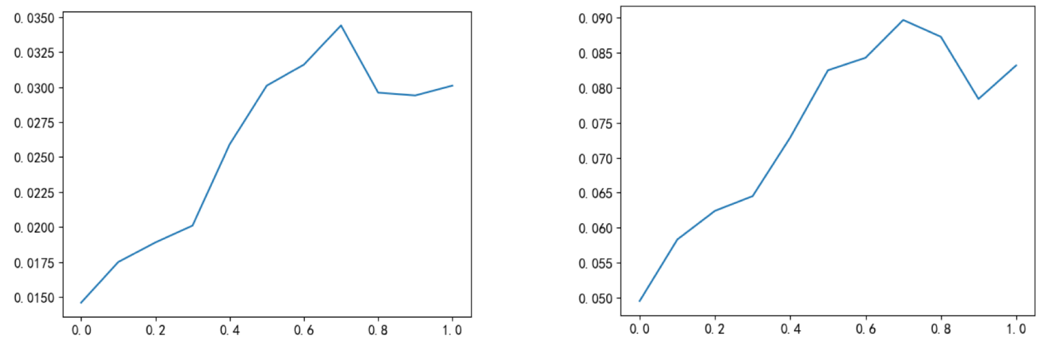
With the integration of ELBO causal extraction and contrastive learning (as depicted in Figure 3(b)), there is a significant improvement in the spatial distribution of preference vectors for both user groups. The preference vectors of cold-start and warm users become more intermixed, resulting in more uniform clusters. This enhancement illustrates that by incorporating ELBO causal extraction and contrastive learning, the model not only captures users' explicit preferences but also delves into deeper causal relationships and latent similarities. Such advancements effectively model and predict the behaviors of cold-start users, aligning their distribution more closely with warm users.

By comparing the changes between the two subfigures, it becomes evident that incorporating ELBO and contrastive learning significantly enhances the preference distribution for cold-start users. This improvement demonstrates the effectiveness of the method we have proposed.

5.5 Hyper-parameters Sensitivity Analysis (RQ4)

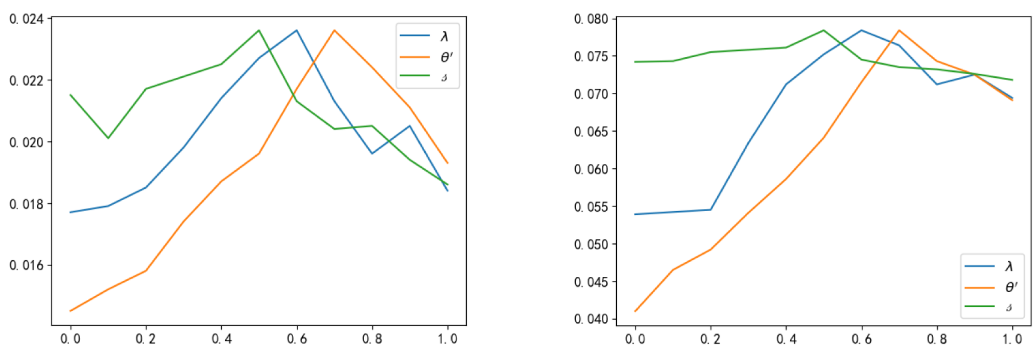
To explore RQ4, this section examines the effects of four hyper-parameters: the ELBO weight for warm users θ, the proportion of long-term user interest $\mathscr{s}$, the ELBO weight for cold-start users θ', and the weight for contrastive learning λ. In the experiment, we adjusted one hyper-parameter at a time, maintaining default values for the others. Due to limitations in space, all four series of experiments were performed on the Epinions dataset, with the results showing trends in HR@K and NDCG@K.

5.5.1 Impact of Warm User ELBO Weight θ. This hyper-parameter primarily controls the weight assigned to causal inference for warm users. Our analysis focuses on the warm user scenario. We performed a sensitivity analysis for θ, exploring values from 0 to 1. As shown in Figure 4, an increase in θ from 0 to 0.7 leads to a progressive improvement in both HR and NDCG, demonstrating



(a) Impact of Parameter θ Variation on HR@20. (b) Impact of Parameter θ Variation on NDCG@20.

Fig. 4. Impact of the ELBO weight θ for warm users.



(a) Impact of hyperparameters Variation on HR@20. (b) Impact of hyperparameters Variation on NDCG@20.

Fig. 5. Impact of hyper-parameters ζ, θ', and λ.

a consistent enhancement in model performance. This indicates that an appropriate ELBO weight can effectively regularize the latent space, improving the accuracy of extracting and recommending based on users' latent preferences. However, further increases in θ result in a decline in HR and NDCG, particularly NDCG, which decreases from 0.0897 to 0.0873 and then further to 0.0784. This suggests that excessive emphasis on extracting latent preferences may detract from the quality of recommendations.

5.5.2 Impact of User Long-term Interest Ratio ζ. This analysis is conducted within the cold-start user scenario, employing discrete values of ζ from 0 to 1 to perform a hyper-parameter sensitivity analysis. This parameter determines the relative weight of long-term interests to item similarity when predicting cold-start user preferences. As shown in Figure 5, when ζ increases from 0 to 0.5, the model's HR and NDCG performance metrics gradually improve, reaching optimal values (HR = 0.0236, NDCG = 0.0784). This indicates that at this ratio, the model effectively combines users' long-term interests with item similarity, thus accurately extracting the candidate primary causal variables for users. However, further increases in ζ beyond 0.5 result in a decline in performance, likely due to the model's excessive dependency on long-term interests and insufficient consideration of the effects of actual user interactions, which leads to less precise extraction of primary causal variables.

5.5.3 Impact of Cold-start User ELBO Weight θ'. This parameter analysis takes place within the context of cold-start user scenarios, employing discrete values of θ' from 0 to 1 for hyper-parameter sensitivity analysis. As illustrated in Figure 5, incrementing θ' from 0 to 0.7 results in gradual improvements in HR and NDCG. This trend suggests that enhancing the ELBO weight aids the model in more effectively extracting hidden user preferences from the characteristics of cold-start users and their social interactions, thus improving the precision and quality of recommendations. However, when θ' exceeds 0.7, HR and NDCG start to decline, likely due to the model overemphasizing the extraction of latent preferences, which may lead to overfitting and inaccuracies in predicting hidden user preferences.

5.5.4 Impact of Contrastive Learning Weights λ. This analysis is performed within the context of cold-start user scenarios, utilizing discrete λ values from 0 to 1 to conduct a hyper-parameter sensitivity analysis. As depicted in Figure 5, a gradual increase in λ from 0 to 0.6 leads to significant improvements in both HR and NDCG. This demonstrates that appropriately increasing the weight of contrastive learning can effectively reduce the distribution gap between cold-start and warm users, thus enhancing the model's accuracy and the quality of its rankings. However, as λ exceeds 0.6, a noticeable decline in performance occurs, particularly evident when λ reaches 0.7. This decline is likely due to the model excessively focusing on contrastive learning, which results in overfitting. Such overfitting makes the distribution of cold-start users overly similar to that of warm users, adversely affecting the extraction of cold-start users' latent preferences.

6 Conclusion

In this article, we explore the cold-start problem in recommender systems. Cold-start users, who lack historical behavioral data, challenge conventional recommender systems in accurately predicting their preferences. To tackle this problem, we propose C2lRec, a method that integrates causal representation and counterfactual inference with social networks to enhance the representations of cold-start users. This approach not only provides more accurate and generalizable recommendations but also adds interpretability to the recommendation process by identifying the key interactions from users' historical behaviors that significantly influence their preferences. Extensive experiments on three real-world public datasets confirm that C2lRec improves recommendation performance for cold-start users. Moreover, as the model also employs causal extraction for warm users, it correspondingly improves the recommendation accuracy in warm user scenarios. Ablation studies corroborate the effectiveness of each component within C2lRec.

However, C2lRec does not perform ideally in recommending cold-start users in scenarios lacking social data. Moreover, our research lacks the extraction of more modal data, such as user comment data, which can significantly enhance user features and effectively improve recommendation results. In the future, we aim to employ large language models to derive variables from user review data, enriching the variables for causal recommendations. We also plan to expand the primary causal variables extraction module, using counterfactual reasoning to augment the interpretability of the recommendation model.

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